Joe Worsfold

Principal Engineer (AI & Platform)

London, United Kingdom · +44 7414 873908 · joseph.worsfold@gmail.com · GitHub · CV repo

Summary

Principal Engineer (AI & Platform) and Technical Director at Stack Foundry Ltd, with 8+ years shipping production systems in insurance and government. Recently delivered a production-scale master data platform for specialty insurance — RAG, embeddings, OpenSearch semantic enrichment, and agentic entity resolution where deterministic matching fails — used by underwriters and dataops. Building agent-to-agent harnesses for local context and document embeddings (Prism), plus Helmsman readiness checks for AI-native engineering. Prior lead roles in Go/TypeScript serverless platforms at Aviva Zero and Flock.

Skills

AI & agents: RAG & embeddings, OpenSearch semantic search, LangChain agentic workflows, LLM selection & evaluation harnesses, Agent-to-agent context assembly, MCP-oriented tool boundaries, Cursor / AI-native engineering, Helmsman AI-readiness checks

Languages: Rust, Go, TypeScript, Python, SQL

Cloud & platform: AWS Lambda, API Gateway, EventBridge, SQS, DynamoDB, ECS, Step Functions, CDK / Terraform, Serverless event-driven architectures

Data: OpenSearch, Neo4j, PostgreSQL, SurrealDB, Pinecone, DynamoDB

Frontend: React, Next.js, HTML / CSS, Accessible UI

Practices: Observability (Datadog, CloudWatch, PagerDuty), CI/CD (GitHub Actions), Tech leadership & mentoring, FCA-aware financial services delivery

Experience

Principal Engineer (AI & Platform)

07/2025 – Present

Stack Foundry Ltd · Remote

Technical Director · sole director

Founder-led engineering consultancy. Principal delivery on AI-native platforms and specialty-insurance data systems; building Prism and Helmsman.

Consultancy started while wrapping Aviva Zero.

Rust, TypeScript, Python, AWS (Lambda, OpenSearch, event-driven), LangChain, RAG/embeddings, SurrealDB, Cursor

- Delivered a production master data management platform for a specialty insurance client in six months, replacing multi-system complexity with one source of truth — fewer engineers to maintain and fewer ops agents to run. Used at production scale by underwriters and dataops.
- Designed LLM selection, RAG, and embedding pipelines with OpenSearch semantic enrichment to improve discoverability across messy insurance master data.
- When deterministic entity resolution failed, introduced LangChain agentic workflows so the system could gather context and propose decisions for human approval — increasing agency without silent autonomous binds.
- Taught client developers AI-native delivery and built bespoke guardrails for agentic coding workflows (Cursor-centred practice).

Senior Full Stack Software Engineer

01/2024 – 11/2025

Aviva Zero · London

Climate-focused insurance platform — high-availability Go/TypeScript services and accessible React frontends.

Backend: Go, AWS Lambda, API Gateway, EventBridge, SQS | Frontend: TypeScript, React, Next.js, ECS | CI/CD: CDK, GitHub Actions, Datadog, PagerDuty

- Designed and implemented event-driven serverless architectures in Go using Lambda, SQS, EventBridge, and SES for asynchronous workflows, scheduling, and alerting across policy and payment services.

- Built and maintained a platform that has sold over a million policies and supports tens of thousands of daily users, with 24/7 reliability through monitoring and performance work.
- Delivered financially sensitive services including BACS payment workflows, interest-bearing monthly instalments, and secure customer interactions within FCA-regulated boundaries.
- Led alerting and observability improvements for proactive detection and resolution.

Lead Software Engineer (Tech Lead) 02/2023 – 01/2024

Flock Cover · London

Next-generation insurance platform with modern cloud and serverless infrastructure.

Backend: TypeScript, Scala, Lambda, API Gateway, EventBridge, SQS | Frontend: React, Next.js, ECS/Lambda | IaC: CDK, Terraform, GitHub Actions, Datadog

- Led full-stack development across Node/TypeScript backends and React/Next.js frontends with a focus on scalability and performance.
- Designed microservices on Lambda, ECS, and EventBridge for observable, 24×7 services that scale with demand.
- Championed Terraform and CDK for repeatable AWS provisioning.

Senior Consultant Full Stack Engineer 10/2021 – 02/2023

Infinity Works (part of Accenture) · UK

Tech lead for specialty-insurance automation and serverless integrations (client work anonymised where required).

TypeScript, Node.js, Postgres, DynamoDB, Lambda, API Gateway, Step Functions, EventBridge, Terraform, Datadog

- Led automation for underwriting data extraction and distribution with TypeScript, Node.js, and Postgres on AWS.
- Built serverless, event-driven integrations with Step Functions and EventBridge for multi-service workflows.
- Acted as Tech Lead — unblocked hard engineering problems, ran knowledge-sharing, and strengthened engineering culture.

Software Engineer 09/2020 – 10/2021

UK Home Office · UK

Graph-backed intelligence tooling and government web applications.

React, Angular, Node.js, MongoDB, Neo4j, Docker, Kubernetes, Drone

- Supported ETL into Neo4j graph databases modelling people, assets, and movements; wrote performant Cypher for pattern discovery at scale — central to the platform's intelligence capabilities.
- Worked with data scientists to operationalise ML against the graph layer for anomaly detection and higher-quality investigative leads.
- Built React/Angular apps and Node REST APIs with TDD/BDD, Docker, and Kubernetes CI via Drone.

IT Developer 09/2018 – 09/2020

Mott MacDonald · UK

ERP integration and process automation.

Java, JavaScript, PL/SQL

- Built Java integrations from legacy systems to cloud ERPs using SOLID, design patterns, and TDD.
- Delivered electronic forms/workflows and PL/SQL-backed data solutions.

Selected projects

Tooling Helmsman

Open rules, skills, and checks for agent-written code — readiness scoring, review, and STAR-stack guardrails. Used on Prism and NIDOS.

Context

Hack Hate 2020: Hate Detector

Winner — Best use of AI/ML (AWS) and Best use of location (ESRI). Processed and classified 18.5M+ tweets; graph model in Neo4j for hate-speech network visualisation.

Presentation · Code

Education

Birkbeck, University of London 2017 – 2019

MSc Computer Science, Distinction

Algorithms, data structures, machine learning, and NLP with emphasis on scalable AI-driven systems.

Thesis: Character-level Convolutional Neural Networks for Hate Speech Detection with Intelligent Adversaries — CNN text classification at scale with graph modelling of classified content in Neo4j.

Royal Holloway, University of London 2009 – 2013

BA History, 2:1

Analytical writing, research, and communication — foundations that still show up in product and stakeholder work.